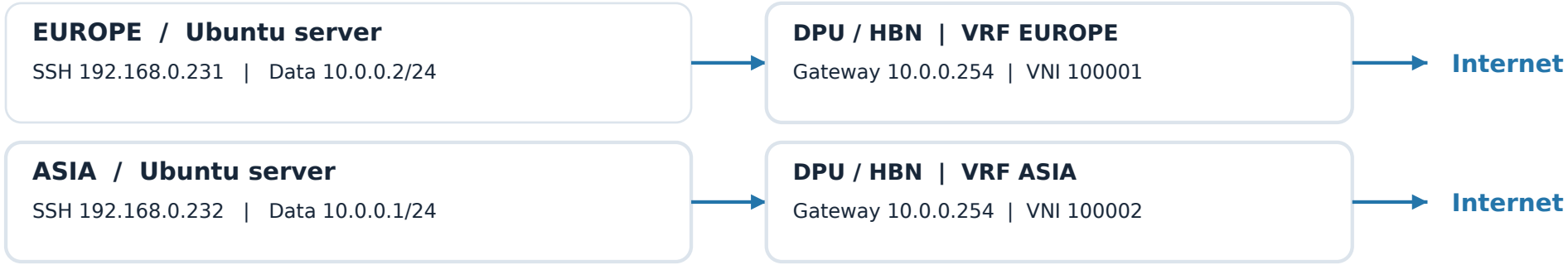
Connect to Europe and Asia

Two Ubuntu servers. Separate HBN tenants. The same IPv4 subnet.

SSH PATH: nico-lab > Kubernetes helper pod > Ubuntu management IP



Different VRFs and VNIs keep the two 10.0.0.0/24 networks isolated.

1 Check readiness on nico-lab

```
kubectl -n dpf-operator-system get dpus
kubectl -n dpf-operator-system get pod dpf-hbn-recovery-access
```

Both DPUs must be Ready and Operational, HBN must be running, and the helper pod must be Ready. A missing Ubuntu data interface means the network demo is not ready.

2 Open Europe SSH from nico-lab

```
ssh -o \
  'ProxyCommand=kubectl -n dpf-operator-system exec -i dpf-hbn-recovery-access -- nc %h %p' \
  ubuntu@192.168.0.231
```

Open Asia SSH in a second nico-lab terminal

```
ssh -o \
  'ProxyCommand=kubectl -n dpf-operator-system exec -i dpf-hbn-recovery-access -- nc %h %p' \
  root@192.168.0.232
```

Use your server SSH key or server password. The BMC password is a different credential.

Show Internet access and tenant isolation

Run these commands inside the Ubuntu SSH sessions from page 1.

3 Confirm the host address and default route

```
DATA_IF=$(ip -4 route show default | awk '/default/ {print $5; exit}')
ip -4 addr show dev "$DATA_IF"; ip -4 route show default
```

Expected: Asia 10.0.0.1/24; Europe 10.0.0.2/24; gateway 10.0.0.254.

4 Prove Internet access through HBN - run on both servers

```
ping -4 -I $DATA_IF -c 3 -W 2 cisco.com
curl --noproxy '*' --interface $DATA_IF --connect-timeout 5 --max-time 15 \
  -sS -o /dev/null -w 'HTTP=%{http_code} source=%{local_ip}\n' https://www.cisco.com
```

Expected: Cisco ping replies and successful HTTPS. The curl source must be 10.0.0.1 for Asia or 10.0.0.2 for Europe. Check that DATA_IF has the expected 10.0.0.x address before continuing.

5 Prove cross-tenant ping is blocked

On Europe: ping Asia

```
ping -4 -I $DATA_IF -c 3 -W 2 10.0.0.1
```

On Asia: ping Europe

```
ping -4 -I $DATA_IF -c 3 -W 2 10.0.0.2
```

Expected: 100% packet loss in both directions, while step 4 succeeds on both hosts. A failed ping alone is not proof: an unfinished installation or a down interface also causes packet loss.

Present it: "Both servers reach the Internet, but their tenant data networks cannot ping each other."

Monitor: tmux attach -t dpf-monitor | Ctrl+b then arrows: select pane | Ctrl+b then z: zoom